Mid-Term Examination, M.Sc. 2nd Semester, Full Marks: 30, Network Security

1. Answer five questions from the following: (5 x 2 = 10)
   a. Prove that $\langle Z_6^*, \times \rangle$ is an abelian group.
   b. List all multiplicative inverse pairs in modulus 20.
   c. Define Avalanche effect.
   d. Is DES a Feistel Cipher? Why or Why not?
   e. Define the term "Statistical Attack". Give an example.
   f. What do you mean by Galois field? How is it related to cryptography?
   g. How a cryptanalysis attack differs from a brute force attack?

2. Answer any five from the following: (5 x 4 = 20)
   a. Discuss, in brief, the difference between Chosen PT attack and Known PT-CT attack through examples.
   b. What is double DES? What kind of attack on double DES makes it useless?
   c. Discuss the advantages of using Counter algorithmic mode for encryption algorithm.
   d. Alice and Bob want to establish a secret Key using Diffie-Hellman Key exchange protocol. The values of n,g,x and y are 11,5,2,3 respectively. Find out the value of the secret key.
   e. Find the GCD of 84 and 320 using Extended Euclidian Algorithm.
   f. The plaintext "letusmeetnow" and the corresponding ciphertext "HBCDFNOPIKLB" are given. You know that the algorithm is a Hill cipher, but you don't know the size of the key. Find the key matrix.